

Rogue Domain Controller

Adversaries may register a rogue Domain Controller to enable manipulation of Active Directory data. DCSshadow may be used to create a rogue Domain Controller (DC). DCSshadow is a method of manipulating Active Directory (AD) data, including objects and schemas, by registering (or reusing an inactive registration) and simulating the behavior of a DC. [1] Once registered, a rogue DC may be able to inject and replicate changes into AD infrastructure for any domain object, including credentials and keys.

Registering a rogue DC involves creating a new server and nTDSDSA objects in the Configuration partition of the AD schema, which requires Administrator privileges (either Domain or local to the DC) or the KRBTGT hash. [2]

This technique may bypass system logging and security monitors such as security information and event management (SIEM) products (since actions taken on a rogue DC may not be reported to these sensors). [1] The technique may also be used to alter and delete replication and other associated metadata to obstruct forensic analysis. Adversaries may also utilize this technique to perform [SID-History Injection](#) and/or manipulate AD objects (such as accounts, access control lists, schemas) to establish backdoors for Persistence. [1]

ID: T1207

Sub-techniques: No sub-techniques

❶

Tactic: [Defense Impairment](#)

❷

Platforms: Windows

Contributors: Vincent Le Toux

Version: 3.0

Created: 18 April 2018

Last Modified: 12 May 2026

[Version Permalink](#)

Procedure Examples

ID	Name	Description
S0002	Mimikatz	Mimikatz 's <code>LSADUMP::DCShadow</code> module can be used to make AD updates by temporarily setting a computer to be a DC. [3][2]

Mitigations

This type of attack technique cannot be easily mitigated with preventive controls since it is based on the abuse of system features.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0276	Detection Strategy for Rogue Domain Controller (DCShadow) Registration and Replication Abuse	AN0770	Detection of rogue Domain Controller registration and Active Directory replication abuse by correlating: (1) creation/modification of nTDSDSA and server objects in the Configuration partition, (2) unexpected usage of Directory Replication Service SPNs (GC/ or E3514235-4B06-11D1-AB04-00C04FC2DCD2), (3) replication RPC calls (DrsAddEntry, DrsReplicaAdd, GetNCChanges) originating from non-DC hosts, and (4) Kerberos authentication by non-DC machines using DRS-related SPNs. These events in combination, especially from hosts outside the Domain Controllers OU, may indicate DCShadow or rogue DC activity.

References

1.

[Delpy, B. & LE TOUX, V. \(n.d.\). DCSshadow. Retrieved March 20, 2018.](#)

2.

[Metcalf, S. \(2015, November 13\). Unofficial Guide to Mimikatz & Credential Theft from Remote Desktops. Retrieved September 29, 2015.](#)

3.

[Deply, B. \(n.d.\). Mimikatz. Retrieved September 29, 2015.](#)